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# Early-lactation diseases and fertility in 2 seasons of calving across US dairy herds

**P. Pinedo,[1]* J. E. P. Santos,[2] R. C. Chebel,[3] K. N. Galvão,[3] G. M. Schuenemann,[4] R. C. Bicalho,[5] R. O. Gilbert,[6] S. Rodriguez Zas,[7] C. M. Seabury,[8] G. Rosa,[9] and W. W. Thatcher[2]**
[1]Department of Animal Sciences, Colorado State University, Fort Collins 80521
[2]Department of Animal Sciences, University of Florida, Gainesville 32611
[3]College of Veterinary Medicine, University of Florida, Gainesville 32611
[4]Department of Veterinary Preventive Medicine, The Ohio State University, Columbus 43210
[5]College of Veterinary Medicine, Cornell University, Ithaca, NY 14850
[6]School of Veterinary Medicine, Ross University, St. Kitts, West Indies
[7]Department of Animal Sciences, University of Illinois, Urbana 61801
[8]College of Veterinary Medicine, Texas A&M University, College Station 77843
[9]Department of Animal Sciences, University of Wisconsin, Madison 53706

## ABSTRACT

The objective of this study was to characterize incidences of health disorders during early lactation in a large population of Holstein cows calving in 2 seasons across multiple US dairy herds. In addition, cumulative effects of combinations of health-related events on fertility and survival by season of calving and parity number were tested. Data were prospectively collected from a total of 11,729 cows in 16 herds located in 2 regions in the United States [north (7,820 cows in 10 herds) and south (3,909 cows in 6 herds)]. Cows were enrolled at parturition and monitored weekly for disease occurrence, reproductive events, and survival. Health-related events were grouped into reproductive disorders (REP; dystocia, twins, retained fetal membranes, metritis, and clinical endometritis) and other disorders (OTH; subclinical ketosis, mastitis, displaced abomasum, and pneumonia). Counts of health events within 50 d postpartum were added into each of the groups and categorized as 0, 1, 2, 3, and ≥4 for REP and 0, 1, 2, and ≥3 for OTH. Multivariable logistic regression was used for testing potential associations between categories of disease occurrence and outcome variables, including resumption of ovarian cyclicity, pregnancy per artificial insemination (AI), pregnancy loss, and survival up to and after 50 DIM. The incidence of disease varied with season of calving and parity, and these 2 variables were associated with the reproductive and survival outcomes. The size of the detrimental effect of disease incidence on reproduction and survival depended on disease group and varied for each specific outcome. Resumption of ovarian cyclicity decreased as incidences of disorders increased in both REP and OTH categories. Pregnancy at first AI also was smaller in greater number of REP categories, but the effect of number of OTH categories on pregnancy at first AI was not consistent. Similarly, pregnancy loss at first AI was not affected consistently by REP or OTH. Survival was reduced by REP and OTH. The magnitude of these negative effects was variable, depending on season of calving and parity, but consistently increased with the number of health events during early lactation.
**Key words:** disease, transition, fertility, multistate

## INTRODUCTION

The risk for suboptimal health is amplified during the transition period, as negative energy balance and immune suppression occur concurrently, leaving cows susceptible to infectious diseases and metabolic disorders (Bell et al., 1995; Esposito et al., 2014). Disease prevalence in early lactation ranges from 30 to 50% in high-producing individuals (LeBlanc et al., 2006, 2010), with the most relevant disorders affecting the reproductive system and the metabolic balance of the cow.

Incidences of these disorders and their relationships with risk factors existing around calving have been described (Nordlund and Cook, 2004; Ingvartsen, 2006), and the detrimental effect of suboptimal health in this period is recognized widely (Santos et al., 2010; Ribeiro et al., 2013). As a result of disease during early lactation, cow performance is reduced and risks of death and culling are increased (de Vries et al., 2010). Moreover, diseases predispose cows to other fertility stressors, such as anovulation and loss of body condition (Santos et al., 2010; Ribeiro et al., 2016).

Although there is consensus on the negative association between diseases and reduced pregnancy per AI

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*Corresponding author: pinedop@colostate.edu

(**PAI**) or extended intervals to pregnancy, the biological mechanisms by which postpartum diseases impair reproduction remain unclear (Ribeiro et al., 2016). The magnitude of the detrimental effects of uterine versus nonuterine disorders on fertility have not been completely documented. In addition, the role of accompanying environmental stressors, such as season of calving and geographic region, on the size of these underlying fertility associations has not been explored extensively.

Notably, the detrimental effects of disease conditions occurring during early lactation extend beyond this period, affecting subsequent biological processes, such as conception and embryo survival (Carvalho et al., 2019). Evidence on the carryover effects of diseases was presented in a series of studies reported by Ribeiro et al. (2016). Inflammatory disease before AI reduced fertilization of oocytes and impaired the development of the early conceptus as well as the maternal recognition process in the uterine lumen. These changes increased pregnancy loss (**PL**) and reduced pregnancy per breeding.

Epidemiological studies on the effect of disease on fertility are based mostly on data stored in on-farm software, where the lack of consistency in recording and definition of diseases is a significant issue. Large multistate prospective studies that use standardized disease definitions and consistent times of assessment to analyze the effect of health events on reproduction are scarce.

Our hypothesis was that incidences of disease during early lactation would be variable across season of calving and parity number. In addition, it was hypothesized that the association among collective occurrence of suboptimal health events and subsequent fertility and survival may be influenced by factors such as season of calving, parity number, and geographic region. Therefore, our objective was to characterize the incidences of health disorders during early lactation in a large population of Holstein cows calving in 2 seasons across multiple US dairy herds using a prospective approach in data collection. In addition, the cumulative effects of combinations of health-related events on fertility and survival by season of calving and parity number were evaluated.

## MATERIALS AND METHODS

All procedures conducted as part of this research were approved by the West Texas A&M University/Cooperative Research, Educational and Extension Team Institutional Animal Care and Use Committee (IACUC; protocol ID: 02-08-12). Farm managers also approved all the procedures.

### *Study Population and Monitoring Procedures*

This prospective study is part of a research effort investigating potential associations between genomic variation and fertility of Holstein cows (Lopes et al., 2020; Pinedo et al., 2020). A convenient sample of 16 herds were enrolled based on existing contacts provided by the researchers, availability of good-quality records, daily recording of milk yield or monthly milk recording service provided by DHI, and geographic distribution across the United States. The study considered a total of 11,733 Holstein cows calving from November 2012 to October 2014 in herds located in 2 regions in the United States [north (7,820 cows in 10 herds in OH, MN, NY, and WI) and south (3,909 cows in 6 herds located in CA, FL, and TX)]. General information regarding participant farms is presented in Table 1.

In each herd, cows were enrolled at parturition, with half of the calvings occurring during a warm season (May to August) or a cool season (October to January) and a targeted inclusion of 33% primiparous cows. Study cows were monitored weekly for disease occurrence, reproductive events, and survival. Collection of individual data started at calving and included farm, date of calving, lactation number, and calving-related events.

Monitoring of disease was performed during weekly visits to the study farms by the authors (i.e., investigator veterinarians and graduate students). Cows were inspected using headlocks or sorting gates after cows returned from the milking parlor. Procedures included the following: (1) evaluation of vaginal discharge at 7 ± 3 and 28 ± 7 DIM by inspection using the Metricheck device (Metricheck, SimcroTech, Hamilton, New Zealand) and a scale of 0 to 5 (0 = no mucus; 1 = crystalline; 2 = flecks of pus; 3 = mucopurulent, <50% pus; 4 = purulent, >50% pus; 5 = watery, reddish or brownish fetid discharge; McDougall et al., 2007); (2) collection of blood samples from the median coccygeal vessels at 7 ± 3 DIM for determination of serum BHB; and (3) lameness scoring at 35 ± 3 DIM as cows returned from the milking parlor using a scale of 1 to 5 [1 = normal; 2 = mildly lame (stands with flat back but arches when walks, gait slightly abnormal); 3 = moderately lame (stands and walk with arched back); 4 = lame (arched back standing and walking, one or more limbs favored); 5 = severely lame (arched back, refuses to bear weight in one limb, great difficulty moving from lying position); modified from Bicalho et al., 2007a].

Events of interest included calving-related events (dystocia, stillbirth, calf sex, and twins; farm records); retained fetal membranes (**RFM**; membranes not expelled after 24 h postcalving; farm records, Kelton et al.,

**Table 1.** General information on the study herds, located in 2 regions (north and south) in the United States[1]

| Location | Cows in milk (no.) | Milkings (no./d) | BTSCC[2] | | Rolling herd milk average (kg) | Heat abatement methods | VWP[3] (DIM) | Synch protocol[4] | Natural Service | Housing for milking cows | Housing for dry cows | Type of milking parlor |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Summer | Winter | | | | | | | | |
| OH | 1,100 | 3 | 340,000 | 240,000 | 13,140 | Fans and sprinklers | 60 | TAI/HD | No | Freestall barns | Freestall barns | Double parallel |
| OH | 1,500 | 3 | 190,000 | 130,000 | 10,585 | Fans and sprinklers | 70 | TAI/HD | No | Freestall barns | Freestall barns | Double herringbone |
| OH | 2,100 | 4 <20 DIM, then 3 | 270,000 | 250,000 | 12,775 | Fans and sprinklers | 60 | TAI/HD | No | Freestall barns | Freestall barns | Double parallel |
| WI | 1,500 | 3 | 300,000 | 285,000 | 14,618 | Fans and sprinklers | 60 | TAI/HD | No | Freestall barns | Freestall barns | Double herringbone |
| MN | 1,900 | 3 | — | — | 16,260 | Fans and sprinklers | 60 | TAI/HD | No | Freestall barns | Freestall barns | Double herringbone |
| WI | 2,700 | 3 | — | — | 14,946 | No | 70 primiparous, 55 multiparous | TAI/HD | No | Cross-ventilated freestall | Freestall barns | Rotary |
| NY | 1,200 | 3 | 310,000 | 280,000 | 14,267 | Fans and sprinklers | 70 | TAI/HD | No | Freestall barns | Freestall barns | Double parallel |
| NY | 958 | 3 | 300,000 | 270,000 | 14,764 | Fans and sprinklers | 50 | TAI/HD | No | Freestall barns | Freestall barns | Double parallel |
| NY | 1,597 | 3 | 200,000 | 165,000 | 13,769 | Fans and sprinklers | 60 | TAI/HD | No | Freestall barns | Freestall barns | Double parallel |
| NY | 2,925 | 3 | 365,000 | 280,000 | 13,271 | Fans and sprinklers | 70 | TAI/HD | No | Freestall barns | Freestall barns | Double parallel |
| TX | 7,150 | 3 | 250,000 | 147,000 | 8,635 | Fans | 50 | No | Yes | Freestall/dry lot | Dry lots with shade | Double parallel |
| TX | 4,194 | 3 | 227,000 | 193,000 | 9,348 | Fans | 50 | TAI/HD | Yes | Freestall/dry lot | Dry lots with shade | Double parallel |
| FL | 5,000 | 3 | 230,000 | 180,000 | 11,300 | Fans and sprinklers | 70 primiparous, 60 multiparous | TAI/HD | No | Freestall barns | Freestall barns | 2 double parallel |
| CA | 1,450 | 2 | 130,000 | 160,000 | 12,500 | Sprinklers | 50 | TAI/HD | No | Dry lots with shades | Dry lots with shade | 2 double herringbone |
| CA | 1,800 | 2 | 150,000 | 140,000 | 12,300 | Sprinklers | 55 | TAI/HD | No | Freestall barns | Dry lots with shade | Double parallel |
| CA | 5,500 | 3 | 130,000 | 120,000 | 13,100 | Fans and sprinklers | 50 | TAI/HD | No | Frestall barn | Freestalls and dry lot | 1 rotary and 1 double herringbone |

[1]North = 7,820 cows in 10 herds in Ohio (OH), Minnesota (MN), New York (NY), and Wisconsin (WI). South = 3,909 cows in 6 herds located in California (CA), Florida (FL), and Texas (TX).

[2]Bulk tank SCC.

[3]Voluntary waiting period.

[4]Synchronization of ovulation protocol. TAI = timed AI; HD = heat detection.

1998), metritis (**MET**; 7 ± 3 DIM; mucus score 5, independent of fever; McDougall et al., 2007), subclinical ketosis (**SCK**; 7 ± 3 DIM; serum BHB >1.0 mmol/L; Overton et al., 2017), clinical endometritis (**END;** 28 ± 7 DIM; mucus score 3 to 4; McDougall et al., 2007), clinical mastitis (**MAS**; 0 = negative or 1 = positive; farm records), left displaced abomasum (**LDA**; farm records), pneumonia (**RESP**; 0 = negative or 1 = positive; farm records with standardized definitions), and lameness (35 ± 3 DIM; score >2; Bicalho et al., 2007a).

Reproductive outcomes included resumption of ovarian cyclicity (**ROC**) that was assessed via transrectal ultrasonography at 40 ± 3 and 54 ± 3 DIM with portable ultrasounds equipped with a 7.5-MHz linear transducer. Anovulation was defined as lack of visible corpus luteum on both consecutive ultrasound scans (Cerri et al., 2004; Bicalho et al., 2008).

Pregnancy diagnosis was performed by ultrasonography on d 32 ± 3 after AI and reconfirmed at d 60 ± 3 of gestation. Pregnancy loss between d 32 and 60 was calculated by the proportion of cows diagnosed pregnant at 32 ± 3 but nonpregnant at 60 ± 3 d after AI.

Additional information included general and production cow data [parity number, milk yield (DHI test day milk yield performed monthly and recorded until 305 DIM), average daily milk yield in the first 90 DIM, season of calving], breeding information (time to first AI and pregnancy, AI per pregnancy), and management-related data (breeding program, voluntary waiting period, and death and live culling).

### Data Organization and Statistical Analyses

A single overall sample size determination was not performed. The multistate approach was to enroll the largest number of cows that was feasible for weekly monitoring to obtain a significant level of variation in multiple fertility variables. Sample size calculations for the analyses of reproductive outcomes and combined death and live culling were performed using the data analysis application SAS Power and Sample Size (release 9.4; SAS Institute Inc., Cary, NC). The calculations considered power = 80% and confidence = 95% and a difference of 3 percentage points between 2 categories of the variables in analysis. The reference values for ROC, PAI, PL, and combined death and live culling were 70, 35, 10, and 5%, respectively. The numbers of cows required to show significant differences for ROC, PAI, PL, and combined death and live culling up to 50 DIM were 2,799, 3,184, 1,398, and 834, respectively.

Data were entered in Microsoft Excel spreadsheets (Microsoft Corp., Redmond, WA), where the information was organized in cow individual records for further analysis using SAS. Calvings were classified as occurring during a warm season (**WS**; May to August) or a cool season (**CS**; October to January), and herd location was assigned to 2 geographic regions in the United States [north (10 herds) and south (6 herds)] to address potential climatic differences (Supplemental Table S1; https://doi.org/10.3168/jds.2019-17951). Cows were categorized by parity number as primiparous (parity 1) or multiparous (parity ≥2).

Frequency distributions (%) for health-related events were calculated by season of calving and parity and by season of calving and region (reported in Table 2). Frequencies (%) for specific reproductive outcomes [ROC, pregnancy at first and second AI (**PAI1** and **PAI2**, respectively), and pregnancy loss at first and second AI (**PL1** and **PL2**, respectively)] and survival were calculated by season of calving, parity, season of calving by parity, region, and season of calving by region (Table 3). The associations among these health-related events and reproduction and survival variables were subsequently tested by logistic regression, with the models including the variables season of calving, parity, and herd as fixed effects.

Univariable analyses were used to test the initial inclusion of each explanatory variable in the final multivariable models exploring potential associations among the study variables. Depending on the analysis, explanatory variables tested for inclusion in the multivariable models included parity category, season of calving, and region. Herd was included in all the models, and average daily milk yield in the first 90 DIM was forced in the analyses of survival up to and after 50 DIM. Variable effects with $P \leq 0.20$ were included in the initial multivariable analyses, and the final models were obtained through a backward elimination procedure. The significance level for inclusion in the final model was set at $P \leq 0.10$ to account for potential confounders affecting both the outcome and the explanatory variables. Depending on the analysis, the interactions between the effect of season of calving, parity, and herd were considered.

Frequency distributions (%) for reproductive and survival outcomes by number of health-related events (reproductive or other ≤50 DIM) were calculated for the complete population (PROQ FREC; Table 4). The associations among season of calving, parity, season of calving by parity, and season of calving by region and the incidence of health-related events were analyzed using the logit link in PROC GLIMMIX for logistic regression (Table 5).

The relationships among multiple levels of the variables in the study and the potential causal associations affecting sequential health disorders, reproductive performance, and combined death and live culling are illustrated in Figure 1. As associations among transi-

**Table 2.** Frequency distributions (%) for health-related events by season of calving[1] by parity category and season of calving by geographic region[2]

| | Parity category | | | | Geographic region | | | |
|---|---|---|---|---|---|---|---|---|
| | Primiparous (n = 4,098) | | Multiparous (n = 7,631) | | North (n = 7,820) | | South (n = 3,909) | |
| Condition[3] | Warm (n = 2,050) | Cool (n = 2,048) | Warm (n = 3,497) | Cool (n = 4,134) | Warm (n = 3,660) | Cool (n = 4,160) | Warm (n = 1,887) | Cool (n = 2,022) |
| Dystocia | 18.5 | 15.8 | 13.3 | 12.0 | 14.3 | 16.0 | 17.1 | 7.76 |
| Stillbirth | 7.34 | 6.80 | 3.85 | 4.20 | 5.70 | 5.68 | 4.08 | 3.76 |
| Twins | 1.00 | 1.20 | 6.30 | 5.60 | 4.48 | 4.90 | 4.08 | 2.40 |
| RFM | 4.49 | 3.71 | 9.84 | 5.85 | 7.62 | 5.58 | 8.32 | 4.30 |
| MET | 32.2 | 34.9 | 21.4 | 20.7 | 21.0 | 21.5 | 33.9 | 33.5 |
| SCK | 25.7 | 13.2 | 33.3 | 18.2 | 31.7 | 16.7 | 28.0 | 16.2 |
| LDA | 0.73 | 0.50 | 3.90 | 2.23 | 3.01 | 1.85 | 2.17 | 1.50 |
| MAS | 13.9 | 10.1 | 12.6 | 12.5 | 14.3 | 11.6 | 10.7 | 11.8 |
| END | 24.6 | 29.9 | 23.0 | 26.3 | 22.6 | 25.7 | 25.6 | 30.9 |
| LAM | 5.63 | 6.51 | 12.6 | 17.3 | 10.5 | 14.9 | 9.00 | 11.4 |
| RESP | 1.01 | 2.44 | 2.60 | 3.30 | 2.00 | 1.50 | 3.55 | 6.30 |

[1]Warm season = May to August; cool season = October to January (years 2012–2014).

[2]North = farms located in New York, Wisconsin, Minnesota, and Ohio; south = farms located in California, Texas, and Florida.

[3]RFM = retained fetal membranes; MET = metritis (7 ± 3 DIM; mucus score 5), SCK = subclinical ketosis (7 ± 3 DIM; serum BHB >1.0 mmol/L); LDA = left displaced abomasum; MAS = mastitis; END = clinical endometritis (28 ± 7 DIM; mucus score >2); LAM = lameness (35 ± 3 DIM; score >2); and RESP = pneumonia.

**Table 3.** Frequency distributions (%) for reproductive outcomes by season of calving, parity category, geographic region, season of calving by parity, and season of calving by geographic region[1]

| Variable | ROC[2] (n = 11,100) | Pregnant AI1 32 d (n = 10,415) | Pregnancy loss AI1 (n = 3,805) | Pregnant AI2 32 d (n = 6,691) | Pregnancy loss AI2 (n = 2,466) | Left herd ≤50 DIM (n = 11,733) | Left herd >50 DIM (n = 11,098) |
|---|---|---|---|---|---|---|---|
| Season of calving[3] | | | | | | | |
| Warm (n = 5,547) | 70.9 | 35.1 | 12.1 | 36.0 | 9.58 | 5.99 | 19.0 |
| Cool (n = 6,182) | 72.0 | 38.2** | 8.3** | 38.0 | 7.88† | 4.90** | 17.4* |
| Parity | | | | | | | |
| Primiparous (n = 4,098) | 69.4 | 40.4 | 8.44 | 37.6 | 7.90 | 3.40 | 10.8 |
| Multiparous (n = 7,631) | 72.6** | 34.7** | 11.1* | 36.7 | 9.11 | 6.50** | 22.1** |
| Parity by season of calving | | | | | | | |
| Primiparous | | | | | | | |
| Warm (n = 2,050) | 69.5 | 40.1 | 10.0 | 36.2 | 8.20 | 3.51 | 10.1 |
| Cool (n = 2,048) | 69.3 | 40.7 | 6.89* | 39.0 | 7.61 | 3.27† | 11.6 |
| Multiparous | | | | | | | |
| Warm (n = 3,497) | 71.8 | 32.0 | 13.9 | 35.9 | 10.4 | 7.43 | 24.2 |
| Cool (n = 4,134) | 73.3† | 36.9** | 9.11** | 37.5 | 8.02* | 5.71** | 20.3** |
| Region[4] | | | | | | | |
| North (n = 7,820) | 70.3 | 36.5 | 8.04 | 36.7 | 7.84 | 5.78 | 18.9 |
| South (n = 3,909) | 74.0** | 37.1 | 13.8** | 37.7 | 10.2* | 4.68* | 16.8** |
| Region by season of calving | | | | | | | |
| North | | | | | | | |
| Warm (n = 3,660) | 70.5 | 35.3 | 9.29 | 36.2 | 7.85 | 6.72 | 18.9 |
| Cool (n = 4,160) | 70.2 | 37.6† | 7.04* | 37.2 | 7.84 | 4.95** | 19.0 |
| South | | | | | | | |
| Warm (n = 1,887) | 71.8 | 34.8 | 17.7 | 35.7 | 12.8 | 4.56 | 19.2 |
| Cool (n = 2,022) | 75.6** | 39.3* | 10.7** | 39.6 | 7.96** | 4.80 | 14.4** |

[1]The significance of the variables effects was tested by logistic regression. Season of calving, parity category, and herd were included as fixed effects in the models.

[2]Resumption of ovarian cyclicity, assessed via transrectal ultrasonography at 40 ± 3 and 54 ± 3 DIM.

[3]Warm season = May to August; cool season = October to January (years 2012–2014).

[4]North = farms located in New York, Wisconsin, Minnesota, and Ohio; south = farms located in California, Texas, and Florida.

†$P < 0.10$; *$P < 0.05$; **$P < 0.01$.

**Table 4.** Frequency distributions (%) for reproductive and survival outcomes by number of health-related events[1] ≤50 DIM presented for the overall population

| No. of health-related events (no. of REP/no. of OTH) | ROC[2] (n = 11,100) | | Pregnant AI1 32 d (n = 10,415) | | Pregnancy loss AI1 (n = 3,805) | | Left herd ≤50 DIM[3] (n = 11,733) | | Left herd >50 DIM (n = 11,098) | |
|---|---|---|---|---|---|---|---|---|---|---|
| | REP | OTH | REP | OTH | REP | OTH | REP | OTH | REP | OTH |
| 0 (5,571/7,534) | 75.1 | 73.7 | 40.5 | 38.1 | 8.46 | 9.81 | 1.33 | 3.58 | 18.1 | 17.2 |
| 1 (3,237/3,284) | 69.4 | 68.4 | 34.1 | 34.8 | 11.0 | 10.6 | 1.61 | 6.70 | 17.8 | 20.1 |
| 2 (1,509/595) | 67.1 | 62.4 | 32.7 | 30.4 | 13.8 | 12.2 | 2.53 | 14.1 | 19.1 | 22.4 |
| 3 (550/41) | 64.1 | 53.1 | 31.3 | 19.2 | 12.4 | 0 | 3.10 | 24.4 | 22.4 | 24.4 |
| ≥4 (205/NA[4]) | 61.6 | — | 23.3 | — | 14.6 | — | 2.94 | — | 24.5 | — |

[1]REP = reproductive events (dystocia, twins, retained fetal membranes, metritis, and clinical endometritis); OTH = other events (subclinical ketosis, mastitis, displaced abomasum, and pneumonia).

[2]Resumption of ovarian cyclicity, assessed via transrectal ultrasonography at 40 ± 3 and 54 ± 3 DIM.

[3]Combined death and live culling before 51 DIM or after 50 DIM.

[4]NA = not applicable.

**Table 5.** Odds ratios (95% CI in parentheses) for the occurrence of health-related events by season of calving, parity, season of calving by parity, and season of calving by geographic region[1]

| | Season of calving[3] | | Parity | | Parity | | | | Geographic region[4] | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Overall (n = 11,729) | | Overall (n = 11,729) | | 1 (n = 4,098) | | ≥2 (n = 7,631) | | North (n = 7,820) | | South (n = 3,909) | |
| Condition[2] | Warm (5,547) | Cool (6,182) | 1 (4,098) | ≥2 (7,631) | Warm (2,050) | Cool (2,048) | Warm (3,497) | Cool (4,134) | Warm (3,660) | Cool (4,160) | Warm (1,887) | Cool (2,022) |
| Dystocia | 1.21 (1.08–1.36)* | — | 1.59 (1.41–1.91)* | — | 1.34 (1.12–1.61)* | — | 1.18 (1.02–1.27)* | — | 0.87 (0.76–0.99)* | — | 2.73 (2.18–3.40)* | — |
| Stillbirth | 1.01 (0.56–1.20) | — | 1.88 (1.59–2.22)* | — | 1.09 (0.85–1.39) | — | 0.96 (0.76–1.21) | — | 0.99 (0.82–1.21) | — | 1.07 (0.88–1.48) | — |
| Twins | 1.13 (0.94–1.40) | — | 0.13 (0.09–0.19)* | — | 1.01 (0.56–2.15) | — | 1.13 (0.93–1.37) | — | 0.97 (0.78–1.20) | — | 1.80 (1.23–2.64)* | — |
| RFM | 1.60 (1.37–1.87)* | — | 0.46 (0.38–0.55)* | — | 1.19 (0.87–1.63) | — | 1.75 (1.47–2.08)* | — | 1.46 (1.22–1.75)* | — | 1.96 (1.48–2.58)* | — |
| MET | 0.95 (0.87–1.04) | — | 1.94 (1.78–2.12)* | — | 0.85 (0.75–0.98)* | — | 1.03 (0.92–1.16) | — | 0.94 (0.84–1.04) | — | 0.98 (0.85–1.13) | — |
| END | 0.80 (0.74–0.88)* | — | 1.17 (1.07–1.28)* | — | 0.73 (0.63–0.85)* | — | 0.86 (0.77–0.95)* | — | 0.84 (0.76–0.94)* | — | 0.75 (0.65–0.87)* | — |
| SCK | 2.28 (2.10–2.50)* | — | 0.63 (0.57–0.69)* | — | 2.54 (2.13–3.04)* | — | 2.40 (2.12–2.66)* | — | 2.47 (2.21–2.77)* | — | 1.95 (1.67–2.29)* | — |
| LDA | 1.79 (1.38–2.31)* | — | 0.17 (0.11–0.26)* | — | 1.57 (0.68–3.61) | — | 1.83 (1.39–2.40)* | — | 1.76 (1.30–2.38)* | — | 1.89 (1.21–3.17)* | — |
| MAS | 1.12 (1.01–1.26)* | — | 0.89 (0.79–1.00) | — | 1.42 (1.16–1.74)* | — | 1.02 (0.88–1.17) | — | 1.22 (1.07–1.40)* | — | 0.93 (0.76–1.14) | — |
| LAM | 0.64 (0.57–0.73)* | — | 0.29 (0.25–0.34)* | — | 0.87 (0.66–1.34) | — | 0.58 (0.50–0.67)* | — | 0.61 (0.52–0.71)* | — | 0.71 (0.57–0.88)* | — |
| RESP | 0.64 (0.50–0.82)* | — | 1.87 (1.41–2.48)* | — | 0.34 (0.19–0.60)* | — | 0.77 (0.58–1.02) | — | 0.88 (0.58–1.32) | — | 0.54 (0.40–0.73)* | — |

[1]Season of calving, parity category, and herd were included as fixed effects in the models.

[2]RFM = retained fetal membranes; MET = metritis (7 ± 3 DIM; mucus score 5); END = clinical endometritis (28 ± 7 DIM; mucus score >2); SCK = subclinical ketosis (7 ± 3 DIM; serum BHB >1.0 mmol/L); LDA = left displaced abomasum; MAS = mastitis; LAM = lameness (35 ± 3 DIM; score >2); and RESP = pneumonia.

[3]Warm season = May to August; cool season = October to January (years 2012–2014).

[4]North = farms located in New York, Wisconsin, Minnesota, and Ohio; south = farms located in California, Texas, and Florida.

*$P < 0.05$.

tion health disorders have been reported (Ingvartsen et al., 2003) and collinearity among variables might affect the outcome of regression models (Schisterman et al., 2017), counts of health-related events were grouped for the analyses, anticipating a cumulative effect of disease on fertility and survival. Therefore, to test the potential effect of incidence of health-related events on reproductive and survival outcomes, health disorders occurring until 50 DIM were grouped into reproductive disorders (**REP**; dystocia, twins, RFM, MET, and END) and other disorders (**OTH**; SCK, MAS, LDA, and RESP). Subsequently, counts for each health event within 50 DIM were added into each of the 2 groups, resulting on the following categories: REP = 0, 1, 2, 3, and ≥4, and OTH = 0, 1, 2, and ≥3. To facilitate the presentation of incidences of health-related events by reproductive outcomes in diagrams, a secondary categorization was performed, grouping the study cows into the following categories: healthy = no health-related event (≤50 DIM); REP 1 = 1 reproductive health event; REP > 1 = 2 or more reproductive health events; OTH 1 = 1 nonreproductive disease event; and OTH > 1 = 2 or more nonreproductive disease events.

Multivariable models using the logit link in PROC GLIMMIX for logistic regression were used to test potential associations among multiple categories of health-related events occurrence (0, 1, 2, 3, and ≥4 for REP and 0, 1, 2, and ≥3 for OTH) and outcome variables, including ROC, PAI1, PL1, and survival (combined death and live culling up to and after 50 DIM; Tables 6 and 7). The analyses were performed for the complete population and separated by season of calving and by parity. Effects were deemed significant when $P \leq 0.05$.

## RESULTS

A total of 11,733 Holstein cows calving from November 2012 to October 2014 in 16 herds located in 2 regions were enrolled in the study and had data available

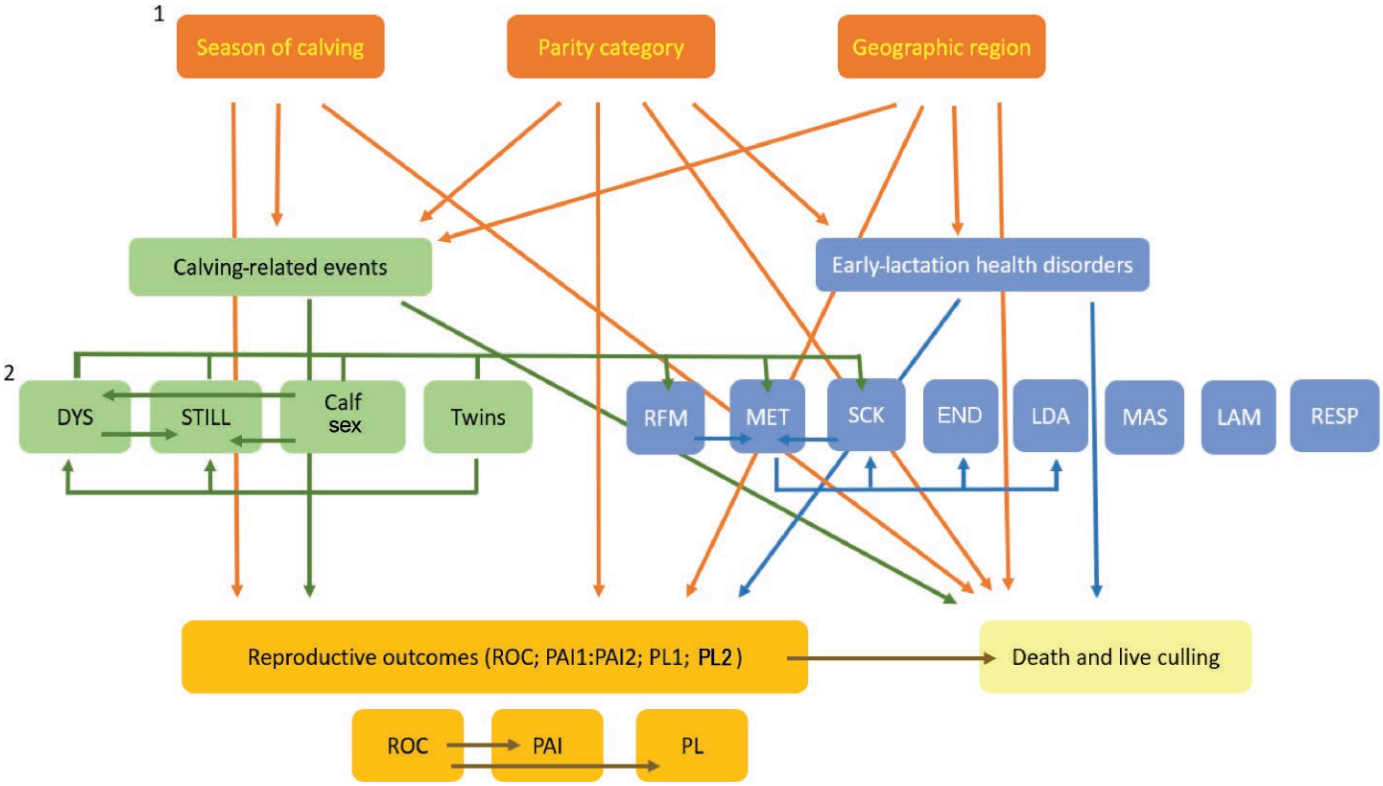


**Figure 1.** Causal diagram illustrating the relationships among multiple levels of the variables in the study and the potential causal associations affecting reproductive performance and combined death and live culling. Arrowheads point to the dependent variables. Season of calving = warm (May to August) and cool (October to January; years 2012–2014). Parity category = primiparous and multiparous. Geographic region = north (farms located in New York, Wisconsin, Minnesota, and Ohio) and south (farms located in California, Texas, and Florida). DYS = dystocia; STILL = stillbirth; RFM = retained fetal membranes; MET = metritis (7 ± 3 DIM; mucus score 5); SCK = subclinical ketosis (7 ± 3 DIM; serum BHB >1.0 mmol/L); END = clinical endometritis (28 ± 7 DIM; mucus score >2); LDA = left displaced abomasum; MAS = mastitis; LAM = lameness (35 ± 3 DIM; score >2); and RESP = pneumonia. ROC = resumption of ovarian cyclicity; PAI1 and PAI2 = pregnancy at first and second AI, respectively; PL1 and PL2 = pregnancy loss at first and second AI, respectively.

**Table 6.** Odds ratios (95% CI in parentheses) for reproductive and survival outcomes by number of health-related events[1] ≤50 DIM presented for the overall population and by season of calving[2]

| Health-related events (no.) | ROC[3] | | Pregnant AI1 32 d | | Pregnancy loss AI1 | | Left herd ≤50 DIM[4] | | Left herd >50 DIM | |
|---|---|---|---|---|---|---|---|---|---|---|
| | REP | OTH | REP | OTH | REP | OTH | REP | OTH | REP | OTH |
| Overall | | | | | | | | | | |
| 0 | — | — | — | — | — | — | — | — | — | — |
| 1 | 0.74 (0.67–0.82)* | 0.81 (0.74–0.90)* | 0.75 (0.68–0.82)* | 0.92 (0.84–1.01) | 1.41 (1.09–1.83)* | 0.98 (0.76–1.27) | 1.18 (0.78–1.78) | 1.93 (1.39–2.68)* | 1.02 (0.90–1.15) | 1.20 (1.07–1.34)* |
| 2 | 0.67 (0.59–0.76)* | 0.63 (0.52–0.77)* | 0.70 (0.62–0.79)* | 0.82 (0.67–1.02) | 1.76 (1.27–2.42)* | 1.23 (0.71–2.12) | 1.01 (0.62–1.65) | 3.74 (2.31–6.05)* | 1.05 (0.90–1.22) | 1.46 (1.18–1.81)* |
| 3 | 0.56 (0.46–0.67)* | 0.43 (0.21–0.89)* | 0.67 (0.55–0.81)* | 0.53 (0.20–1.42) | 1.67 (0.99–2.81) | 1.00 (0.01–99.9) | 1.16 (0.60–2.26) | 2.28 (0.30–17.2) | 1.17 (0.93–1.46) | 1.63 (0.74–3.59) |
| ≥4 | 0.49 (0.36–0.66)* | NA[5] | 0.45 (0.32–0.64)* | NA | 2.32 (0.94–5.71) | NA | 0.42 (0.13–1.30) | NA | 1.10 (0.78–1.56) | NA |
| Warm season | | | | | | | | | | |
| 0 | — | — | — | — | — | — | — | — | — | — |
| 1 | 0.78 (0.67–0.90)* | 0.72 (0.63–0.82)* | 0.80 (0.70–0.93)* | 0.93 (0.82–1.07) | 1.37 (0.96–1.97) | 1.08 (0.77–1.52) | 1.18 (0.65–2.14) | 2.41 (1.53–3.80)* | 0.89 (0.75–1.06) | 1.15 (0.98–1.34) |
| 2 | 0.77 (0.64–0.93)* | 0.51 (0.40–0.66)* | 0.68 (0.56–0.82)* | 0.86 (0.64–1.14) | 1.26 (0.77–2.05) | 1.15 (0.56–2.39) | 0.91 (0.45–1.81) | 6.00 (3.32–10.8)* | 0.95 (0.76–1.19) | 1.42 (1.06–1.89)* |
| 3 | 0.58 (0.44–0.75)* | 0.38 (0.16–0.91)* | 0.67 (0.50–0.88)* | 0.35 (0.08–1.53) | 1.71 (0.83–3.49) | 1.00 (0.01–99.9) | 0.97 (0.40–2.34) | 3.92 (0.49–31.3) | 1.02 (0.75–1.39) | 2.04 (0.79–5.23) |
| ≥4 | 0.55 (0.36–0.83)* | NA | 0.39 (0.23–0.66)* | NA | 2.34 (0.63–8.61) | NA | 0.09 (0.01–0.85)* | NA | 1.25 (0.78–1.99) | NA |
| Cool season | | | | | | | | | | |
| 0 | — | — | — | — | — | — | — | — | — | — |
| 1 | 0.70 (0.61–0.80)* | 0.89 (0.77–1.03) | 0.71 (0.62–0.80)* | 0.89 (0.77–1.02) | 1.55 (1.06–2.26)* | 0.80 (0.52–1.22) | 1.28 (0.70–2.35) | 1.53 (0.92–2.53) | 1.12 (0.95–1.33) | 1.23 (1.05–1.45)* |
| 2 | 0.59 (0.49–0.70)* | 0.78 (0.57–1.06) | 0.73 (0.61–0.86)* | 0.77 (0.56–1.06) | 2.58 (1.66–4.01)* | 1.50 (0.65–3.47) | 1.08 (0.53–2.21) | 1.32 (0.46–3.82) | 1.12 (0.91–1.39) | 1.53 (1.10–2.13)* |
| 3 | 0.52 (0.39–0.69)* | 0.48 (0.14–1.67) | 0.68 (0.51–0.90)* | 0.86 (0.22–3.34) | 1.80 (0.82–3.98) | 1.00 (0.01–99.9) | 1.49 (0.50–4.44) | 1.00 (0.01–99.9) | 1.35 (0.97–1.88) | 0.97 (0.21–4.56) |
| ≥4 | 0.42 (0.27–0.65)* | NA | 0.51 (0.32–0.83)* | NA | 2.27 (0.65–8.00) | NA | 1.14 (0.31–4.23) | NA | 0.95 (0.57–1.60) | NA |

[1]REP = reproductive events (dystocia, twins, retained fetal membranes, metritis, and clinical endometritis); OTH = other events (subclinical ketosis, mastitis, displaced abomasum, and pneumonia).

[2]Season of calving, parity category, and herd were included as fixed effects in the models. Average daily milk yield in the first 90 DIM was included in the analysis associated with survival outcomes. Warm season = May to August; cool season = October to January (years 2012–2014).

[3]Resumption of ovarian cyclicity, assessed via transrectal ultrasonography at 40 ± 3 and 54 ± 3 DIM.

[4]Combined death and live culling before 51 DIM or after 50 DIM.

[5]NA = not applicable.

*$P < 0.05$.

**Table 7.** Odds ratios (95% CI in parentheses) for reproductive and survival outcomes by number of health-related events[1] ≤50 DIM presented by parity category[2]

| Health-related events (no.) | ROC[3] | | Pregnant AI1 32 d | | Pregnancy loss AI1 | | Left herd ≤50 DIM[4] | | Left herd >50 DIM | |
|---|---|---|---|---|---|---|---|---|---|---|
| | REP | OTH | REP | OTH | REP | OTH | REP | OTH | REP | OTH |
| Primiparous | | | | | | | | | | |
| 0 | — | — | — | — | — | — | — | — | — | — |
| 1 | 0.78 (0.66–0.92)* | 0.77 (0.65–0.91)* | 0.68 (0.58–0.80)* | 0.88 (0.74–1.03) | 1.45 (0.93–2.27) | 1.06 (0.67–1.67) | 1.63 (0.65–4.10) | 1.41 (0.72–2.76) | 0.91 (0.70–1.18) | 1.10 (0.86–1.40) |
| 2 | 0.66 (0.54–0.80)* | 0.53 (0.37–0.78)* | 0.62 (0.51–0.75)* | 0.87 (0.59–1.29) | 1.57 (0.92–2.68) | 0.75 (0.22–2.61) | 1.12 (0.40–3.08) | 3.75 (1.41–9.94)* | 1.12 (0.83–1.50) | 0.99 (0.56–1.75) |
| 3 | 0.63 (0.46–0.85)* | 1.02 (0.09–12.0) | 0.66 (0.48–0.90)* | 1.00 (0.01–99.9) | 1.96 (0.89–4.32) | 1.00 (0.01–99.9) | 1.17 (0.29–4.72) | 1.00 (0.01–99.9) | 1.01 (0.64–1.60) | 1.00 (0.01–99.9) |
| ≥4 | 0.59 (0.35–1.01) | NA[5] | 0.45 (0.25–0.82)* | NA | 2.93 (0.76–11.2) | NA | 0.93 (0.09–9.48) | NA | 0.74 (0.32–1.71) | NA |
| Multiparous | | | | | | | | | | |
| 0 | — | — | — | — | — | — | — | — | — | — |
| 1 | 0.74 (0.65–0.83)* | 0.84 (0.74–0.95)* | 0.79 (0.70–0.89)* | 0.93 (0.83–1.05) | 1.39 (1.01–1.92)* | 0.90 (0.65–1.24) | 1.08 (0.66–1.74) | 2.10 (1.43–3.08)* | 1.07 (0.93–1.22) | 1.27 (1.12–1.44)* |
| 2 | 0.71 (0.60–0.85)* | 0.64 (0.51–0.81)* | 0.77 (0.65–0.92)* | 0.78 (0.61–1.00) | 1.89 (1.26–2.84)* | 1.32 (0.71–2.45) | 1.01 (0.57–1.79) | 3.69 (2.09–6.52)* | 1.01 (0.84–1.21) | 1.66 (1.31–2.11)* |
| 3 | 0.52 (0.41–0.67)* | 0.39 (0.18–0.85)* | 0.67 (0.51–0.88)* | 0.31 (0.09–1.05) | 1.48 (0.73–3.03) | 1.00 (0.01–99.9) | 1.15 (0.53–2.51) | 2.68 (0.35–20.5) | 1.21 (0.93–1.56) | 1.84 (0.82–4.12) |
| ≥4 | 0.47 (0.33–0.67)* | NA | 0.44 (0.28–0.69)* | NA | 2.16 (0.62–7.54) | NA | 0.36 (0.10–1.27) | NA | 1.21 (0.82–1.77) | NA |

[1]REP = reproductive events (dystocia, twins, retained fetal membranes, metritis, and clinical endometritis); OTH = other events (subclinical ketosis, mastitis, displaced abomasum, and pneumonia).

[2]Season of calving and herd were included as fixed effects in the models. Average daily milk yield in the first 90 DIM was included in the analysis associated with survival outcomes.

[3]Resumption of ovarian cyclicity, assessed via transrectal ultrasonography at 40 ± 3 and 54 ± 3 DIM.

[4]Combined death and live culling before 51 DIM or after 50 DIM.

[5]NA = not applicable.

*$P < 0.05$.

for the analyses. The frequencies (%) for reproductive health events and other diseases by season of calving by parity and season of calving by geographic region are presented in Table 2. Results from the multivariable logistic regression analyses indicated variation for these outcomes depending on season of calving and parity category (Table 5). Except for mastitis, parity had a significant effect on all the heath-related events. As shown in Table 5, some of these associations changed when the analyses were completed by calving season by parity and by calving season by region.

For example, dystocia occurred more frequently in primiparous cows, but both parity groups experienced a greater incidence of dystocia in the WS. The occurrence of a WS increase in dystocia was more prominent in the south region. Overall, the frequency of twins was less in primiparous cows and twinning was most common in the south during WS. Associated with the increased WS occurrence of dystocia was an increase in RFM, which occurred at a higher frequency for multiparous cows than for primiparous cows. Warm season of calving was also associated with increased incidences of SCK, LDA, and MAS, whereas incidences of END, lameness, and RESP were greater during CS. Primiparous cows also had greater incidence of stillbirth, MET, END, and RESP. On the contrary, incidences of RFM, SCK, LDA, and lameness were smaller in this group of cows.

The observed frequency distributions for the reproductive (ROC, PAI, and PL) and survival (combined death and live culling before 51 DIM or after 50 DIM) outcomes in analysis by season of calving, parity, season of calving by parity, region, and season of calving by region are presented in Table 3. Cows calving during the CS had greater PAI1 and smaller PL at AI1 and AI2. However, no significance was established for the differences in ROC and PAI2 between CS and WS.

The proportion of cows leaving the herd was smaller in CS (up to and after 50 DIM). The significance of these associations remained for multiparous cows but was only partial for primiparous animals. Likewise, when the analyses were performed by region, the effect of season was more notable in cows in the south subgroup, where ROC and PAI1 were greater during CS and PL1, PL2, and death and culling were smaller in CS.

Reproductive and survival outcomes were associated with parity. Resumption of ovarian cyclicity by 54 d postpartum was greater in multiparous cows, with the increase being most notable in the CS of the south region. Pregnancy at AI1 was lower in multiparous cows that were affected negatively by the WS compared with primiparous cows. Lower pregnancy at AI1 during the WS was evident for both the north and south regions. Although pregnancy losses at AI1 were higher for multiparous cows, losses were always lower for CS in both primiparous and multiparous cows in both the north and south regions. As expected, frequency of culling was greater during the period >50 DIM than during the period ≤50 DIM and was greater for multiparous cows. Culling for both periods was higher in the north region.

Overall frequency distributions for the reproductive and survival outcomes categorized by number of health-related events (0, 1, 2, 3, ≥4) are presented in Table 4. Frequencies for ROC, PAI1, and PAI2 consistently decreased with greater categories of both reproductive and nonreproductive health-related events. In general, PL and the number of cows leaving the herd increased in greater disease categories. The odds ratios for reproductive and survival outcomes by category of REP and OTH health-related events in the overall population and divided by season of calving are provided in Table 6. Resumption of ovarian cyclicity was affected consistently by REP; the odds of resuming ovarian cyclicity for cows with REP health events in categories 1, 2, 3, and ≥4 were 0.74, 0.67, 0.56, and 0.49 times the odds, respectively, of ROC for cows with no health-related events (REP category 0). Similarly, the odds of ROC for cows in OTH categories 1, 2, and ≥3 were 0.81, 0.63, and 0.43 times the odds, respectively, of ROC for REP category 0 (i.e., cows without OTH events).

Likewise, categories of REP (1, 2, 3, and ≥4 compared with no health-related events) negatively influenced the odds of PAI1; the odds of pregnancy were 0.75, 0.70, 0.67, and 0.45 times the odds, respectively, of PAI1 for cows with no health-related events (REP category 0). In contrast, no significant effects of OTH disease categories on PAI1 pregnancy were detected.

Pregnancy loss at first AI was increased partially by REP; the odds of PL for cows in REP categories 1 and 2 were 1.41 and 1.76 times the odds, respectively, of PL for cows in REP category 0. In contrast, PL was not affected by OTH events. The odds of leaving the herd up to 50 DIM were greater only in OTH categories 1 and 2 compared with category 0 (odds ratio = 1.93 and 3.74, respectively). Similarly, the odds of leaving the herd after 50 DIM were greater only in OTH categories 1 and 2 compared with category 0 (odds ratio = 1.20 and 1.46, respectively).

Most of the significant associations indicated for the overall population remained when cows were divided by season of calving (Table 6). However, a loss of significance was established for the effect of OTH on ROC during CS, for REP on PL in WS, and for OTH on survival ≤50 DIM during CS. Odds of culling at ≥50

DIM for greater OTH events occurred across both WS and CS, whereas increased odds of culling at ≤50 DIM for increased OTH events were mainly in the WS.

Table 7 reports the results for the logistic regression analyses testing the association between reproductive and survival outcomes and REP and OTH health events partitioned by parity categories. In both primiparous and multiparous cows, greater categories of REP and OTH were in general associated with smaller odds of ROC compared with cows with no health-related events (REP and OTH categories 0). Greater categories of REP were associated with smaller PAI1 values in primiparous and multiparous cows, whereas an association between OTH and PAI1 was not established. Neither PL nor survival exhibited consistent associations with REP or OTH. However, multiparous cows in OTH categories 1 and 2 had greater odds of leaving the herd both up to and after 50 DIM, and this appeared to be associated with multiparous cows in REP categories having greater odds of PL to AI1.

When all the study cows were combined and grouped by season of calving (WS vs. CS), as illustrated in Figure 2, both REP and OTH had significant detrimental effects on ROC, PAI1, and maintenance of pregnancy. However, effects were variable in magnitude and direction among seasons of calving and region (Figure 3).

The detrimental effect of health-related events on ROC was evident in both regions but more consistent for calvings during the WS (Figure 3). Similarly, a trend for lower PAI1 as health-related events category increased was visible in both seasons of calving. Healthy cows had higher PAI1 in the CS. When PL was considered, there was not a clear pattern for the effect of health-related events on maintenance of pregnancy. However, overall PL was greater in the south during the WS.

## DISCUSSION

The detrimental effect of peripartal disease on fertility is recognized widely. However, large multistate studies that test these associations using a prospective approach with standardized disease definitions and consistent times of assessment are scarce. Our hypotheses were based on the concept that the effect of stressors during the transition period would vary depending on warm or cool conditions and would affect primiparous and multiparous cows differently.

The prospective study of 16 farms grouped into north and south regions of the United States documented the seasonal (warm vs. cool) potential cumulative effects of health events in early lactation on reproductive performance and culling of primiparous and multiparous Holstein cows. The incidence rates of the health-related events considered in our study (Table 2) were within the ranges previously reported (Ingvartsen, 2006; LeBlanc et al., 2006; LeBlanc, 2010; Vergara et al., 2014). As expected, some of the disorders in our study population varied significantly among season of calving and parity (i.e., RFM, END, and SCK). In addition, as observed in disorders such as RFM and SCK, the magnitude of the differences between seasons was partially affected by region (Tables 2 and 5).

Although standardized definitions for the health disorders were implemented in this investigation, the criteria to determine a specific disorder may differ among published studies. An example is SCK, whereby different cut-point values for serum BHB concentrations

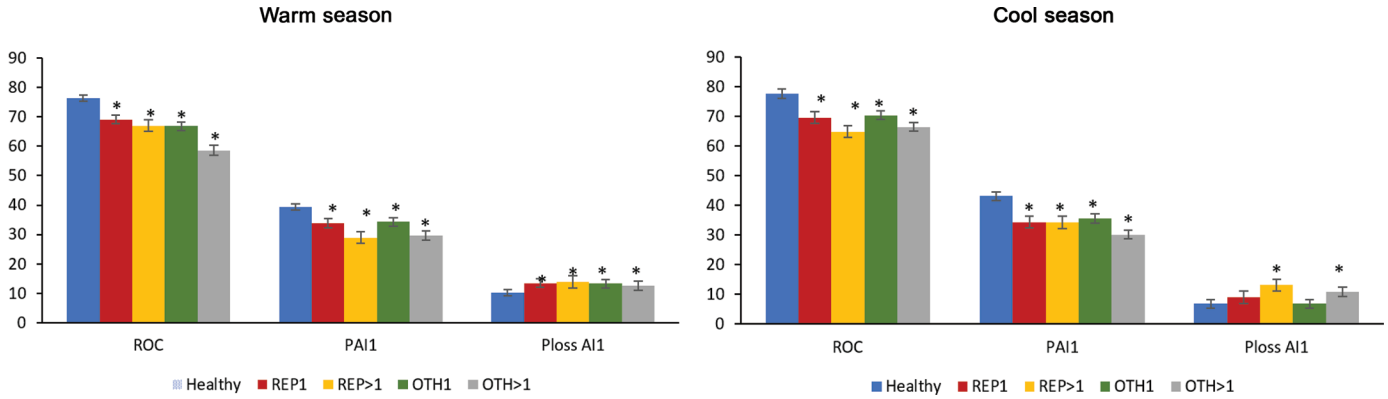


**Figure 2.** Frequencies (%) of reproductive outcomes [resumption of ovarian cyclicity (ROC), pregnancy at AI1 (PAI1), and pregnancy loss at AI (PlossAI1)] by number of disease events (≤50 DIM) for the overall population. Healthy = no disease event (≤50 DIM); REP1 = 1 reproductive disease event (dystocia, twins, retained fetal membranes, metritis, and clinical endometritis); REP>1 = 2 or more reproductive disease events; OTH1 = 1 non-reproductive disease event (subclinical ketosis, mastitis, displaced abomasum, and pneumonia); OTH>1 = 2 or more non-reproductive disease events. Warm season = May to August; cool season = October to January (years 2012–2014). Error bars represent SEM. $^*P < 0.05$.

were used depending on the detrimental effect under consideration (Overton et al., 2017). Another point to consider is the level of subjectivity in the assessment of a specific disease as well as the origin of data (i.e., respiratory disease reported by the farm). However, in this study, most of the disease information was prospectively collected by the authors. Thus, an adequate control of variation due to reporting bias was expected due to consistency in disease definitions and time of assessment across the herds. As illustrated in Figure 1, another element adding complexity to this study was the relationships between specific health events, which could affect disease incidence, as well as the magnitude of the detrimental effects on reproductive performance and death and live culling.

In agreement with our results across geographical regions and seasons, significant differences in disease incidence across herds were presented by Dubuc and Denis-Robichaud (2017) when analyzing 126 commercial dairy herds in Canada. A wide range of herd prevalence was reported for purulent vaginal discharge (0 to 45%), cytological endometritis (5.3 to 80%), and leukocyte esterase endometritis (0 to 78%). Similarly, prevalence of hyperketonemia, displaced abomasum, and retained placenta ranged from 4.0 to 75%, 0 to 11.5%, and 0.0 to 16.7%, respectively.

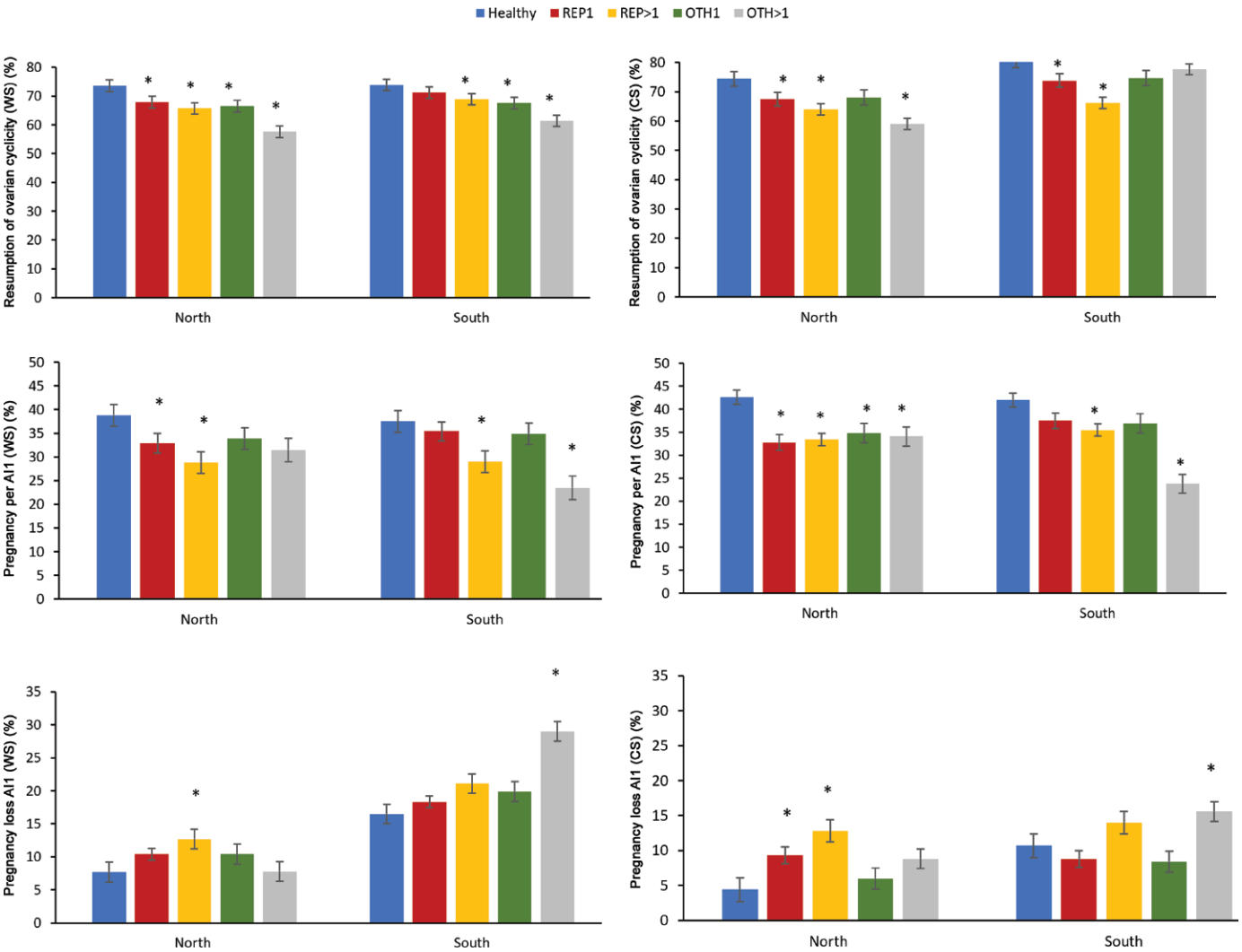


**Figure 3.** Frequencies (%) of reproductive outcomes (resumption of ovarian cyclicity, pregnancy at AI1, and pregnancy loss at AI1) by number of disease events (≤50 DIM) presented by geographic region within season of calving. Healthy = no disease event (≤50 DIM); REP1 = 1 reproductive disease event (dystocia, twins, retained fetal membranes, metritis, and clinical endometritis); REP>1 = 2 or more reproductive disease events; OTH1 = 1 non-reproductive disease event (subclinical ketosis, mastitis, displaced abomasum, and pneumonia); OTH>1 = 2 or more non-reproductive disease events. WS = warm season (May to August); CS = cool season (October to January; years 2012–2014). North = farms located in New York, Wisconsin, Minnesota, and Ohio; south = farms located in California, Texas, and Florida. Error bars represent SEM. *$P < 0.05$.

In the present study, MET incidence was an example of geographic variation (Table 2). Incidences of 33.9 and 33.5% in the south during WS and CS, respectively, were higher than values determined in the north (21.0 and 21.5% in WS and CS, respectively). Conclusions regarding regional effects are somewhat tentative due to the nature of convenience selection of farms for these extensive and intensive investigations. Nevertheless, distinct increases in occurrences of RFM, SCK, and LDA in the north and south were detected during the WS.

In agreement with the reported levels of variation in disease incidence, some of the odds ratios for occurrence of diseases by season of calving, parity, season of calving by parity, and season of calving by region were indicative of differential effects (Table 5). Essentially, incidence of multiple diseases and culling differed between seasons. Although the magnitude of the calculated odds ratios was moderate, the most extreme examples for seasonal variation included metabolic diseases, such as SCK [odds ratio (95% CI) = 2.28 (2.10–2.50) for WS vs. CS] and LDA [odds ratio = 1.79 (1.38–2.31) for WS vs. CS], that occurred with greater frequency in the WS. Even though occurrences of dystocia and RFM were greater in the WS, the postpartum uterus appears to be more resistant to this effect with no seasonal difference in occurrence of MET and a decrease in occurrence of END in the WS (Tables 2 and 5). This is an interesting dynamic relationship in the WS that was detected for END in both the north and south regions. Lewis et al. (1984) demonstrated that cows provided with no shade during late gestation in June had a greater percentage of their uteri restored to the pelvic canal at earlier times postpartum than did cows housed in the shade prepartum. Cows in no shade prepartum had higher concentrations of 13,14-dihydro-15-keto-prostaglandin from the day of parturition until about 16 d postpartum. Environmental effects during late gestation appear to affect postpartum uterine function and competence.

As anticipated, the association between parity number and incidence of health-related events also was significant for most of the disorders in the study (Table 5). The odds of dystocia, stillbirth, MET, and END were greater in primiparous cows, which agrees with previous reports (Kinsel et al., 1998; Bicalho et al., 2007b). In contrast, occurrence of twins, RFM, SCK, LDA, and lameness was greater in multiparous cows (Table 5), which agrees with previous research (LeBlanc et al., 2002; Bicalho et al., 2007b; Dubuc et al., 2010; Vieira-Neto et al., 2017). Also, lower parity number was a protective factor against death and live culling, as previously reported by de Vries et al. (2010). The proportion of primiparous cows leaving the herd up to and after 50 DIM was about half that of multiparous cows (Table 3). Overall, greater proportions of cows left the herd during the WS. This trend was consistent for both primiparous and multiparous cows leaving the herd <50 DIM. When cows were analyzed grouped by region by calving season, more cows in the north left the herd ≤50 DIM in the WS, whereas more cows in the south left the herd at >50 DIM in summer (Table 3).

The overall ROC frequency close to 71% by 50 DIM (Table 3) is comparable with the 64.8% median that occurred at 50 ± 7 DIM in Canadian dairies (Dubuc and Denis-Robichaud, 2017). The ROC frequency from the present study agreed with a 20-herd study (77.5%; Walsh et al., 2007) and 2 herd studies of 77.8 and 73.8% (Dubuc et al., 2012; Vieira-Neto et al., 2014) conducted at similar times during the postpartum period. Moreover, other studies have reported similar ranges for cows remaining anovulatory by 50 to 60 DIM (Butler, 2003; Santos et al., 2009; Galvão et al., 2010). In agreement with previous reports, when ROC was analyzed by parity category (parity 1 vs. parity ≥2) there was a difference in favor of multiparous cows (72.6 vs. 69.4%). Santos et al. (2009) reported lower levels of ROC for primiparous compared with multiparous cows, which may be explained by greater incidence of MET (Goshen and Shpigel, 2006) as well as a more severe negative energy balance (Wathes et al., 2007) in these cows.

Frequency distributions of reproductive outcomes varied appreciably by season of calving and parity categories. The lower reproductive performance of multiparous cows consistently improved for those calving in the CS (Table 3). Beneficial reproductive differences due to the cool calving season were evident in both the north and south geographical regions. Greater overall pregnancy rates at 32 d for AI1 in primiparous cows (40.4%) than in multiparous cows (34.7%) are consistent with those of Vieira-Neto et al. (2017). When season of calving and parity were considered, pregnancy per AI1 in the present population ranged from 32.0% (multiparous, WS) to 40.7% (primiparous, CS; Table 3), and high temperature-humidity index during summer could partially explain this variation.

Greater PAI1 in CS was evident in both the north and south regions. However, the detrimental effect of the WS on PAI1 was most evident in multiparous cows. Indeed, greater losses of pregnancy to AI1 followed the same pattern of pregnancy at AI1 of multiparous cows being more sensitive than primiparous cows to the WS. Seasonal increases in losses of pregnancy to AI1 occurred in both north and south regions. Furthermore, no differences in PAI were established for AI2 in cows that calved in WS that may have progressed to milder conditions at the time of their second AI and of course

later prepartum (Table 3). The pregnancy responses of the present study agree with studies performing pregnancy diagnosis between 30 and 35 d (Santos et al., 2009; Galvão et al., 2010; Vieira-Neto et al., 2014).

Pregnancy loss is significant in dairy cows, and the reported prevalence varied from 6 to 39% (Chebel et al., 2004; Moore et al., 2005; Hernandez et al., 2012). However, differences in time of diagnosis should be considered when comparing different studies. Pregnancy loss in the present study was determined between d 30 and 60 of pregnancy. As described by Wiltbank et al. (2016), the time between d 28 and 60 after AI is a critical period. Factors indicated as responsible include delays or defects in development of the chorioallantoic placentomes or embryo resulting in corpus luteum regression or embryo death. Some of the key developmental problems are inadequate placentation, difficulty in the transition from amniotic to allantoic nutrition, alterations in vascularization of placenta and other systems, and underdevelopment of the embryo or fetus.

In the current study, when cows were grouped by parity and calving season, PL for AI1 was smallest in primiparous cows calving in CS (6.89%), whereas the greatest PL was determined in multiparous cows in WS (13.9%). Overall, PL1 was greater in multiparous cows than in primiparous cows (11.1% vs. 8.44%), which agrees with findings reported previously (Humblot, 2001; Vieira-Neto et al., 2017). Indeed, greater losses of pregnancy to AI1 followed the same pattern of pregnancy at AI1, with multiparous cows being more sensitive than primiparous cows to the WS. Seasonal increases in losses of pregnancy to AI1 occurred in both north and south regions, with the greatest PL levels in the south subgroup calving in WS.

These values are in the lower range compared with those reported in 2 studies in California that used similar assessment dates (Chebel et al., 2004; Moore et al., 2005) and indicated values of 13% (from d 30 to 45 after AI) and 18% (28 d after AI in cows diagnosed on d 27). Similarly, in the review by Wiltbank et al. (2016) that summarized 46 studies with diagnosis times at 32 and 60 d after AI, losses of 12% from more than 20,000 pregnancies in high-producing dairy cows were reported, which is comparable with the 13.9% for multiparous cows of the present study.

Summer heat is a significant stressor affecting dairy cows in the United States (St-Pierre et al., 2003), and direct negative effects on milk yield, health, and reproduction have been well documented (West, 2003; de Vries and Risco, 2005). Consequently, it is expected that the present population of lactating dairy cows would experience lower reproductive performance during WS, especially in areas affected by high temperature and humidity during summer.

Although the present prospective experiment detected seasonal variation for the analyzed reproductive outcomes, the direction of the differences was not consistent. In the case of ROC, the greatest values occurred in the south during CS. However, no differences were determined in the north group of herds. Similarly, in the case of PAI1, values in the south in CS were the greatest. Interestingly, herds in the north also performed better during CS (37.6%) compared with WS (35.3%; Table 3). In agreement with previous data (López-Gatius et al., 2006), pregnancy loss (AI1) was consistently greater during WS in all the subgroups reported in Table 3. Notably, the magnitude of the differences between season of calving was the greatest for WS versus CS in the south.

The detrimental effects of postpartum suboptimal reproductive health, especially uterine-related disorders, on fertility also are well recognized (Dubuc et al., 2011; Gilbert, 2019). Gröhn and Rajala-Schultz (2000) indicated that cows experiencing RFM had a 14% lower conception rate than unaffected cows. Extended days to first service, reduced conception rates at first service, and increased days to conception together with higher risk for culling have been reported for cows affected by MET (Fourichon et al., 2000; Gröhn et al., 2003). Similarly, END resulted in prolonged time to pregnancy, reduced likelihood of conception, and increased risk for PL (Bonnett et al., 1993; Galvão et al., 2009; Bicalho et al., 2016).

Alternatively, health disorders in other nonreproductive systems have also been related to worsened fertility. Cows experiencing postpartum hyperketonemia are at a greater risk of facing poor reproductive performance (Walsh et al., 2007; Ospina et al., 2010; McArt et al., 2012), including PL at first service (Santos et al., 2004, 2009). Similarly, Ribeiro et al. (2013) determined that cows considered to be healthy were more likely to be cyclic than cows with at least 1 clinical or subclinical disease. In addition, cows with clinical and subclinical disease had a reduced probability of being cyclic compared with cows with either a clinical or subclinical disease only. In that study, healthy cows had greater PAI on d 30 and 65 after insemination than cows diagnosed with diseases, and cows with diagnosis of multiple diseases had greater PL than cows with no or a single clinical disease.

Ribeiro et al. (2016) and Carvalho et al. (2019) documented that the detrimental effect of disease conditions extends beyond early lactation, affecting subsequent biological processes such as conception and embryo survival. Moreover, Ribeiro et al. (2016) highlighted the carryover consequences of diseases occurring before breeding on developmental biology, with potential effects lasting up to 4 mo.

In the present study, there was a consistent trend for a reduction in the odds of ROC and pregnancy at AI1 as the number of health-related episodes increased. As shown in Tables 4 and 6, this trend was very consistent for both REP and OTH disorders. However, contrary to data from Ribeiro et al. (2016) and Carvalho et al. (2019), when the effects of health-related events on PL were considered, the only significant associations occurred for 1 and 2 REP events both overall and in CS compared with healthy cows. No health effects on PL1 were detected for any of the OTH categories (Table 6). Interestingly, when cows were grouped by parity or season of calving categories, the significant differences on PL were again reported for 1 and 2 REP events versus healthy multiparous cows (Table 7), and such losses due to 1 and 2 REP occurred in the CS but were not as great in the WS. This may be due to pregnancies initiated at AI1 in multiparous cows being more sensitive to PL. However, number of cows in REP categories 3 and 4 was too few to explore further differences in sensitivity of PL to health events.

To better describe the effect of the 2 groups of health-related events on the reproductive outcomes of interest, REP and OTH were categorized as 1 and >1 and contrasted with cows that did not have any health-related event (Figures 1 and 2). The graphic plots showed some variation in magnitude and direction when cows were grouped by region and season. The detrimental effect of health-related events on ROC was evident in most of the groups, with the exemption of OTH during CS in the south. A consistent trend for lower PAI1 as health-related events increased was most evident for REP in the north in both seasons. However, the significance of the associations was not consistent for REP in the south for either season and for OTH in both regions. When PL was considered, there was not a clear pattern for the significance of the effects of disease on maintenance of pregnancy, except for REP in the north during CS. It was clear that overall PL1 in the south for the WS was substantially greater than the cold season.

Overall, cows that had 1 or 2 metabolic health issues postcalving (i.e., non-reproductive health issues) had greater odds of leaving the herd due to death or culling either before or after 50 DIM (Table 6). This trend was consistent with culling in multiparous cows experiencing 1 and 2 metabolic events compared with healthy cows (Table 7).

Odds of being culled or dying were not associated with the early event of first AI. Producers are somewhat forgiving at that stage of early lactation regarding culling of cows that do not conceive to first service compared with cows that fail to conceive at subsequent services. On the contrary, the effect of repeated non-reproductive health-related events (OTH) by itself could be expected to increase subsequent culling at either early or later stages of lactation. However, as reported previously, the consequent failure to conceive or remain pregnant beyond first AI is also a major reason why cows are culled (Bascom and Young, 1998; Hadley et al., 2006). Nevertheless, pregnancy status beyond second AI for cows that failed to conceive was not considered in this study. In agreement, de Vries et al. (2010) reported that pregnant cows had 3 to 7 times lower hazards of culling than open cows. Hazards of culling were increased for cows that had greater calving difficulty, gave birth to males or twins, were in herds with shorter days to first insemination, or had longer days to conception.

Although the size of the study population in the current report was considerable (n = 11,733), the reduction in the number of cows per category resulting from stratification by parity, calving season, and geographic region affected our statistical power for detecting significant differences in some subcategories with low frequencies. Moreover, some outcomes, such as PAI2 and PL2, included only the fraction of cows that did not conceive or did not maintain pregnancy after AI1. As an example, to be able to determine as significant a difference in PL2 between primiparous and multiparous cows as large as that of PL1 [2.66 percentage points (Table 3) with power = 80% and confidence = 95%], a total sample size of 3,080 cows would be required. However, our number of available cows for this calculation was 2,466 cows.

## CONCLUSIONS

Incidences of health-related events during early lactation were variable depending on season of calving and parity number. Resumption of ovarian cyclicity and pregnancy at first AI were affected consistently by the occurrence of reproductive health events (REP) and non-reproductive health events (OTH) during early lactation. The magnitude of these effects depended on season of calving and parity number but increased with the number of health-related events.

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## ORCIDS

P. Pinedo https://orcid.org/0000-0001-7111-3377
J. E. P. Santos https://orcid.org/0000-0003-3403-1465
R. C. Chebel https://orcid.org/0000-0002-9700-8089
K. N. Galvão https://orcid.org/0000-0002-6683-4476
G. M. Schuenemann https://orcid.org/0000-0003-1114-4512
R. C. Bicalho https://orcid.org/0000-0003-4680-1981
R. O. Gilbert https://orcid.org/0000-0002-1528-243X
G. Rosa https://orcid.org/0000-0001-9172-6461
W. W. Thatcher https://orcid.org/0000-0001-6866-3431